```
(*This is out full expression withouth the prefactor*)
```

In[169]:=

$$\text{Sin}\Delta\phi 2\text{Sin}\theta 22\text{Sin}\theta 12 = \text{sin1}^\wedge 2\ \text{sin2}^\wedge 2 \left(1 - \frac{\left(-2\,k\,q\,(1+q)^2\sqrt{r} + \left(2\,c2\,q^3\sqrt{r}\,\chi 2 + 2\,c1\,q\,\chi 1\left(\sqrt{r} + c2\,q\,\chi 2\right) + \left(\chi 1^2 + q^4\,\chi 2^2\right)\right)\right)^2}{4\ \text{sin1}^\wedge 2\ \text{sin2}^\wedge 2\ M^5\,q^4\,\chi 1^2\,\chi 2^2}\right)$$

Out[169]=

$$\left(1 - c1^2\right)\left(1 - c2^2\right)\left(1 - \frac{\left(-2\,k\,q\,(1+q)^2\sqrt{r} + \chi 1^2 + 2\,c2\,q^3\sqrt{r}\,\chi 2 + q^4\,\chi 2^2 + 2\,c1\,q\,\chi 1\left(\sqrt{r} + c2\,q\,\chi 2\right)\right)^2}{4\left(1 - c1^2\right)\left(1 - c2^2\right)q^4\,\chi 1^2\,\chi 2^2}\right)$$

```
(*The sines can go away from the denominator*)
```

In[170]:=

$$\text{Taylor} = \text{sin1}^\wedge 2\ \text{sin2}^\wedge 2 - \frac{1}{4\,M^5\,q^4\,\chi 1^2\,\chi 2^2}\left(-2\,k\,q\,(1+q)^2\sqrt{r} + M^2\left(2\,c2\,q^3\sqrt{r}\,\chi 2 + 2\,c1\,q\,\chi 1\left(\sqrt{r} + c2\sqrt{M}\,q\,\chi 2\right) + \sqrt{M}\left(\chi 1^2 + q^4\,\chi 2^2\right)\right)\right)^2$$

Out[170]=

$$\left(1 - c1^2\right)\left(1 - c2^2\right) - \frac{\left(-2\,k\,q\,(1+q)^2\sqrt{r} + \chi 1^2 + 2\,c2\,q^3\sqrt{r}\,\chi 2 + q^4\,\chi 2^2 + 2\,c1\,q\,\chi 1\left(\sqrt{r} + c2\,q\,\chi 2\right)\right)^2}{4\,q^4\,\chi 1^2\,\chi 2^2}$$

```
(*From previous derivation we know the expression of the c1=
 cosθ1 and c2=cosθ2 as a function of δχ and χcons*)

M = 1;
```

In[171]:=

```
c1new = FullSimplify[
  PowerExpand[1/((2 + k1 + k2) M q χ1 χ2) (2 Sqrt[M] q (1 + q) Sqrt[r] χ2 + (1 + k2) M q (1 + q) δχs χ2 -
      Sqrt[(M q^2 (1 + q)^2 (c^2 (2 + k1 + k2) M - 2 c (2 + k1 + k2) Sqrt[M] Sqrt[r] + 4 r + 2 (-k1 + k2)
            Sqrt[M] Sqrt[r] δχs + (1 - k1 k2) M δχs^2) χ2^2)]) /. {c → χcons, δχs → δχ}]]
c2new = FullSimplify[PowerExpand[
  -1/((2 + k1 + k2) M q^2 χ2^2) (-2 Sqrt[M] q (1 + q) Sqrt[r] χ2 + (1 + k1) M q (1 + q) δχs χ2 +
      Sqrt[(M q^2 (1 + q)^2 (c^2 (2 + k1 + k2) M - 2 c (2 + k1 + k2) Sqrt[M] Sqrt[r] + 4 r + 2 (-k1 + k2)
            Sqrt[M] Sqrt[r] δχs + (1 - k1 k2) M δχs^2) χ2^2)]) /. {c → χcons, δχs → δχ}]]
```

Out[171]=

$$\frac{1}{(2+k1+k2)\,\chi1}(1+q)\left(2\sqrt{r}+\delta\chi+k2\,\delta\chi-\sqrt{4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi\text{cons}^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi\text{cons})}\right)$$

Out[172]=

$$-\frac{1}{(2+k1+k2)\,q\,\chi2}(1+q)\left(-2\sqrt{r}+\delta\chi+k1\,\delta\chi+\sqrt{4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi\text{cons}^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi\text{cons})}\right)$$

```
(*this is the final expression*)
```

In[173]:=

```
Taylorsimp = FullSimplify@
  PowerExpand[Taylor /. {χeff → χeffdef, δχ → δχdef} /. {c1 → c1new, c2 → c2new}]
```

Out[173]=

$$\left(1-\frac{1}{(2+k1+k2)^2\,\chi1^2}(1+q)^2\left(2\sqrt{r}+\delta\chi+k2\,\delta\chi-\sqrt{\left(4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi cons^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi cons)\right)}\right)^2\right)$$

$$\left(1-\frac{1}{(2+k1+k2)^2\,q^2\,\chi2^2}(1+q)^2\left(-2\sqrt{r}+\delta\chi+k1\,\delta\chi+\sqrt{\left(4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi cons^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi cons)\right)}\right)^2\right)-$$

$$\frac{1}{4\,q^4\,\chi1^2\,\chi2^2}\left(-2\,k\,q\,(1+q)^2\sqrt{r}+\chi1^2+q^4\,\chi2^2-\frac{1}{2+k1+k2}2\,q^2\,(1+q)\sqrt{r}\left(-2\sqrt{r}+\delta\chi+k1\,\delta\chi+\sqrt{\left(4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi cons^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi cons)\right)}\right)-\right.$$

$$\frac{1}{(2+k1+k2)^2}2\,q\,(1+q)\left(2\sqrt{r}+\delta\chi+k2\,\delta\chi-\sqrt{\left(4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi cons^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi cons)\right)}\right)$$

$$\left.\left(-\left((4+k1+k2+2\,q)\sqrt{r}\right)+(1+q)\left(\delta\chi+k1\,\delta\chi+\sqrt{\left(4\,r+(1-k1\,k2)\,\delta\chi^2+(2+k1+k2)\,\chi cons^2-2\sqrt{r}\,((k1-k2)\,\delta\chi+(2+k1+k2)\,\chi cons)\right)}\right)\right)\right)^2$$